Acta Astronautica 167 (2020) 280–285

Contents lists available at ScienceDirect

Acta Astronautica

journal homepage: www.elsevier.com/locate/actaastro

Research paper

Does artificial intelligence dream of non-terrestrial techno-signatures?

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ABSTRACT

Today, we live in the midst of a surge in the use of artificial intelligence in many scientific and technological applications, including the Search for Extraterrestrial Intelligence (SETI). However, human perception and decision-making is still the last part of the chain in any data analysis or interpretation of results or outcomes. One of the potential applications of artificial intelligence is not only to assist in big data analysis but to help to discern possible artificiality or oddities in patterns of either radio signals, megastructures or techno-signatures in general. In this study, we review the comparative results of an experiment based on geometric patterns reconnaissance and a perception task, performed by 163 human volunteers and an artificial intelligence convolutional neural network (CNN) computer vision model. To test the model, we used an image of the famous bright spots on the Occator crater on Ceres. We wanted to investigate how the search for techno-signatures or oddities might be influenced by our cognitive skills and consciousness, and whether artificial intelligence could help or not in this task. This article also discusses how unintentional human cognitive bias might affect the search for extraterrestrial intelligence and techno-signatures compared with artificial intelligence models, and how such artificial intelligence models might perform in this type of task. We discuss how searching for unexpected, irregular features might prevent us from detecting other nearside or in-plain-sight rare and unexpected signs. The results strikingly showed that a CNN trained to detect triangles and squares scored positive hits on these two geometric shapes as some humans did.

1. Introduction

In its exobiology funding programme aims for 2019, NASA has included a non-radio techno-signature option, while China has built the biggest radio telescope on Earth. However, the old questions still remain, as does the silence. The SETI strategy, primarily based on radio signals, has been unsuccessful for decades. Although the goal and spirit remain in good shape it is the approach itself that may be suffering from tunnel vision, otherwise known as the cosmic gorilla effect [1]. In the 1990s, psychologists Simon and Chabris from Harvard popularised an experiment in which half the observers missed a man in a gorilla costume crossing the scene because they were busy counting how many ball passes men in white t-shirts performed. This was due to what is now called the inattentional blindness effect [2]. Our anthropomorphic or somehow still *Ptolemaic* scientific view of the cosmos is due to our brain structure, senses, evolution and mind. We can only grasp a very limited portion of physical reality, and the scale of the universe is mind-challenging itself. If we agree that science has evolved dramatically since the late '50s and '60s when the radio signal approach was conceived, now—in a digital, quantum era—maybe it is time for us to update this view with other possibilities. Our concept of what an extraterrestrial is could be wrong. We suppose that 'they' have to travel vast distances, send radio, laser beams or burst space probes through the immense cosmos. However, other human mind-challenging possibilities may exist. Independently of the methods or technology they may use, we could focus on the traces, techno-signatures or bio-signatures that they might leave behind. We propose that they could leave behind signs or forms of techno-signatures other than radio signals that we might be able to detect if we look in the right direction and pay attention. Although this may sound obvious, it can become really tricky and confusing.

We humans conceive and model reality to fit our own convenience, experience and concepts—extraterrestrial intelligence being no exception. Hollywood movies influence our lives and shape our view of extraterrestrial civilisations and intelligence, from the pet-like ET to militarised Star Wars and predatory aliens, all very naïve and humane in essence. Frequently when we, including scientists, talk about extraterrestrials, we tend to see them as somehow akin either to us or to robots, using radio waves and numbers, sending blueprints as an act of goodwill or even living around Dyson sphere-like [3] megastructures. The truth could be quite different. More advanced civilisations are perhaps simply incomprehensible to us. Known intelligent civilisation classifications to date consider energy consumption as a key factor. Such classifications may represent a short-range approach to the problem. Most probably, advanced civilisations will be beyond our technologically comprehensible horizon, may dominate dark matter/unknown energies and may be multidimensional but we really don't know.

The fact is that silence in our spectrum persists. Among the reasons for this silence we can mention an array of factors, including 1) the wrong technological approach on our side, (2) human brain/consciousness and (3) 'their' nature and intentions [1]. It is interesting to note that to date some efforts and attention have been directed at factor

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https://doi.org/10.1016/j.actaastro.2019.11.013
Received 22 August 2019; Received in revised form 6 October 2019; Accepted 9 November 2019
Available online 15 November 2019
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1, but less or none at factors 2 and 3. For example, in the new nomenclature proposed for SETI, an old term, techno-signature, has been brought back, which involves the detection of radio signals, lasers, atmospheric pollution, radiation leakage from megastructures or sidereal installations such as Dyson spheres, Shkadov thrusters [4] with the power to alter the orbits of stars around the Galactic Center, etc. Some authors have postulated the possibility of previous ancient civilisations indigenous to our solar system [5] having left behind some techno-signatures that we might find. However, if we look for these techno-signatures, artificial structures or signs, our minds can easily become confused when confronted with the unexpected. The question is whether our minds are ready and capable of finding and understanding such techno-signatures, or whether we need to wait for our consciousness to be able to apprehend and comprehend these phenomena. In the meantime, perhaps, we could get some help with this task from artificial intelligence.

Searching for unexpected infrequent elements may prevent us from detecting other nearside infrequent unexpected signs; or as Aristotle put it, "'persons do not perceive what is brought before their eyes, if they are at the time in deep thought, or in a fright, or listening to some loud noise". A real possibility is that factor 2, human brain functioning or our level of consciousness, is limiting our search and our understanding of the universe and more advanced intelligent life living within it. Our understanding of reality is limited and its determined by the circuitry of our brains, and this put us in a difficult position when confronted with the unknown and unexpected, such as another advanced intelligence or other cosmological aspects. Factor 3 is also relevant because some advanced intelligences may simply prefer to remain undetected without renouncing to interact. This strategy could be interpreted as a form of ecological/naturalistic research approach as we sometimes do when interacting with other species in nature. Another possibility is intentional avoidance.

Artificial intelligence (AI) models in various scientific fields have been used to improve prediction power and forecasting accuracy over older methods. Within these AI models, computer vision helps on tasks such as pattern recognition and the classification and reclassification of events or images, etc. Computer vision model applications include Medicine, Agriculture, Safety and Astronomy [6,7] In this paper, for our first aim we focused on factor 2 to test our consciousness and cognitive *modus operandi* compared with AI, by way of a visual perception experiment using a Convolutional Neural Network (CNN) AI computer vision model.

A second aim of this study was to test whether a trained AI CNN model could help to discover new patterns where such patterns may have been overlooked, and to categorise and classify the data free of human influence and cognitive limitations in the least biased way possible. Possible applications from this type of study could be new AI algorithms resulting in standardised tools and methodologies that could be applicable to different types of techno-signature search (radio, image, etc.). These AI models could provide probabilistic data about what type of signal or pattern might possibly be detected and its potential artificiality characteristics.

2. Methods

A comparative visual perception and pattern recognition experiment was performed by humans and by an AI CNN model of computer vision. For the experiment we used a section of the NASA Dawn probe image PIA21925 (Fig. 1), specifically a section from the Vinalia Faculae region in Ceres's Occator crater, i.e. part of the popular bright spots. This particular image was chosen for two main reasons: first because of the interest of astronomy and the general public in the bright spots and their possible origin, making it a good, controversial candidate to elucidate; and second, because in a pre-pooled selection of planetary images, including several from Mars and Ceres, many volunteers perceived some geometrical patterns, particularly squared patterns, in it. In

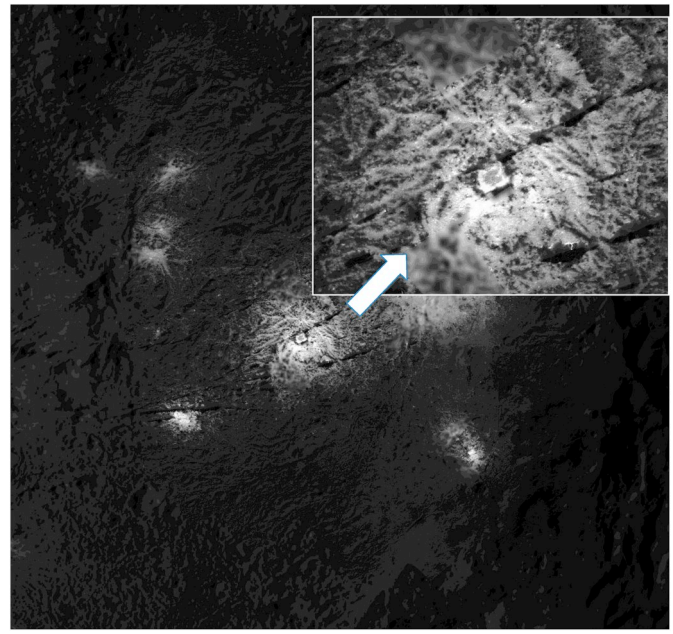


Fig. 1. PIA 21,925 image, Occator crater, Ceres. NASA Dawn Probe. Inset upper right: section image from original PIA21925 used for the experiment. Original image credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA/PSI.

a popular NASA online questionnaire surveying general public opinion on the origin of the bright spots, 10% of respondents opted for volcano, 10% geyser, 6% rock, 30% ice and 8% salt deposit, while 38% opted for '*other*', before the Dawn spacecraft arrived and closer views of the formations on Occator could be studied. The more likely scientific option was salt formation.

According to NASA, this 'image was obtained by NASA's Dawn spacecraft in its second extended mission, from an altitude as low as 21 miles (34 km). The contrast in resolution obtained by the two phases is visible, reflected by a few gaps in the high-resolution coverage (blurry parts). This image is superposed to an equivalent scene acquired in the low-altitude mapping orbit of the mission from an altitude of about 240 miles (385 km)' (https://solarsystem.nasa.gov/resources/1095/mosaic-of-the-vinalia-faculae-in-occator-crater/).

2.1. Human perception task

For the purpose of the task testing human performance, we recruited a sample of 163 participants. All participants were older adults, 40 men and 123 women with a mean age of 22.29 years (SD 3.48), and all were volunteers. None of them had training in astronomy, a related specialty or expert satellite imagery analysis. The task consisted of three stages. In first stage, an example was given to the participants in which they were shown a satellite picture with some clear geometric forms on it (square) and asked to draw over any geometric patterns they thought they could detect. After reading the instructions they had to turn over the page to where the image PIA 2195 section of Vinalia Faculae (Occator crater on Ceres) was shown (Fig. 2). Once they felt they had completed that task they could go on to page 3, where two questions had to be answered: one of the questions asked whether they had detected a big triangle pattern in the picture (a traced image including this triangle was shown (see Fig. 2)); another was whether they believed they could now make out the traced triangle after not having done so previously.

2.2. Convolutional Neural Network Model

The computer vision model used to analyse the image was an AI

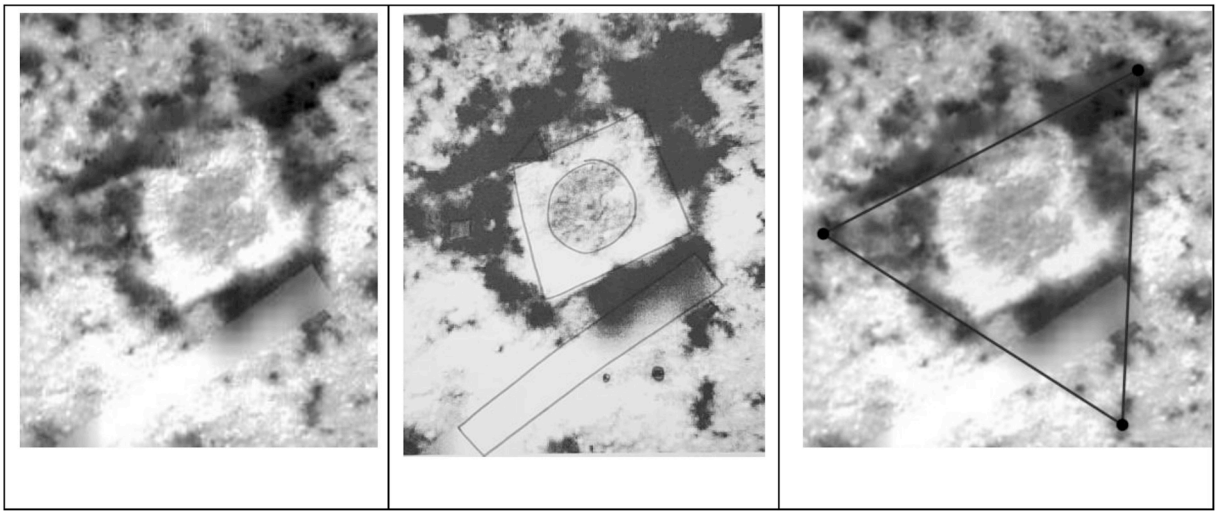


Fig. 2. Test image (section of PIA21925) (left), example of participant's response (center) and traced image for final questionnaire (right). Original image credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA/PSI.

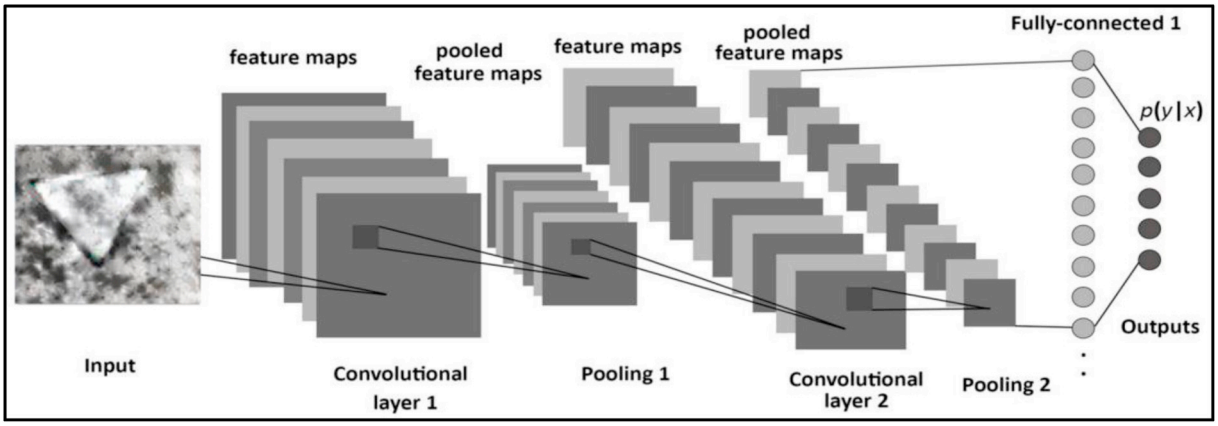


Fig. 3. Convolutional Neural Network Model (CNN) architecture used in the study. In this study, we focused on images containing geometric shapes. Specifically, the database employed contained 10,000 images (Fig. 4), both colour and gray-scale.

Fig. 4. Sample images used for the training of the Artificial Intelligence (AI) computer vision of Convolutional Neural Network Model (CNN). Upper row: Images with geometric properties including artificially generated images representing shapes in satellite imagery. Lower row: background images, non-targeted geometric figures and unrelated imagery.

model based on Convolutional Neural Networks (CNNs) (Fig. 3) [8], a type of computer vision model based on deep learning [9]. CNNs are a category of neural networks that have been successful in areas such as image recognition and classification. Some examples of their application include the recognition of faces, objects and traffic lights. In addition, they have revolutionised robotic vision systems and autonomous vehicles. CNNs have to be trained in order to equip them with adaptation skills. During training, the CNN learns by using a set of representative images that depict the different objects to be classified or detected. The following categories were used:

- Artificially generated images representing shapes without borders.
- Artificially generated images representing shapes with borders.
- Artificially generated images representing shapes with distorted borders, simulating freehand drawings.
- Artificially generated images displaying only a background.
- Natural images displaying different objects, obtained from the VOC 2007 dataset [10] and unrelated.
- Artificially generated images representing shapes simulating satellite imagery.

Since we wanted to analyse images obtained from space probes, for the CNN to obtain accurate results we needed similar images in the training database. However, it is obviously difficult to obtain real images that include geometric shapes. For this reason, we adapted the rest of the artificially generated images using a CNN able to combine an input image with a style image [11]. During the training of the model we applied data augmentation techniques. Specifically, the images could be rotated up to 45° and flipped horizontally and vertically, and could suffer small variations in lighting and zoom. As for the architecture of the CNN, we used the pre-trained ResNet 34 network model with the weights obtained for the ImageNet dataset [12]. We fine-tuned the weights using fast.ai and pytorch libraries. The final validation accuracy obtained was 99.49% (Fig. 5). To classify a new image, we used the test-time augmentation technique, which aggregates n predictions obtained using data augmentation on the new image. Specifically, we used n = 20.

3. Results

In Fig. 7 we can see the five patterns most frequently recognised by our human participants. Feature 1 was the pattern most frequently recognised, as predicted, because it represented a well-defined rectangular area in which the space probe Dawn did not get the best resolution conforming to a rectangle section in the image. This pattern was considered our control stimulus because it was obviously artificial. It was followed in popularity by pattern 3, which was perceived to be a circle. Pattern 5 (the big, darker triangle) was the least often detected feature in the first instance. However, after participants performed the reconnaissance task, *a posteriori* when they were asked whether they could see a real triangle after it was traced, the percentage detecting it rose from 7.1% to 56% (Table 1). The five patterns most frequently detected and perceived by participants conform to an interesting overall figure that is very interrelated geometrically (Fig. 6).

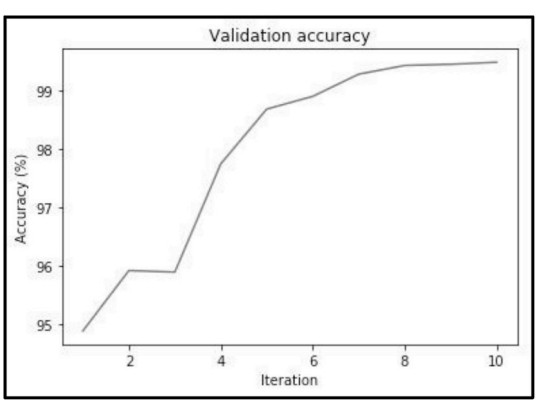


Fig. 5. Validation accuracy curve for trained model.

Table 1
Recognition percentages for the most frequent patterns perceived by human participants. 1: Big rectangular formation, left center. 2: Big square formation, center. 3:circular shape, center of square 3. 4: small squared formation, up right. 5A: Big Triangle before perception test, 5B: Big Triangle after perception test.

Recognition	5A	5B	1	2	3	4
yes	11.00	63.20	85.30	47.20	66.90	36.20
no	89.00	36.80	14.70	52.80	33.10	63.80

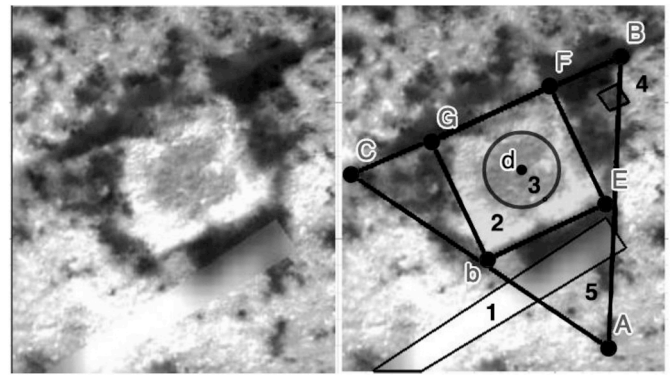


Fig. 6. Left: section of PIA2195 used for the test. Right: Most frequent patterns perceived by human participants in the reconnaissance task. Original image credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA/PSI.

As for the CNN AI model, the results are strikingly similar to those obtained by the humans for the images of VF-1. However, the CNN AI model was also tested with two other Dawn images for contrast and observation of the model's performance. To our surprise, the model still detected both triangle and square formations or patterns on images PIA22626 at 58 km of altitude and PIA20653 at 385 km, both images having been obtained by Dawn at earlier stages of the Ceres exploration (Table 2). There is a constant detection result for both triangular and square patterns. This confirms that the CNN computer vision AI model detected at least two patterns compatible with two different geometric forms (triangle and square) in the same formation in three different images of the same location at Vinalia Faculae, Occator crater, the bright region on Ceres.

Older imaging of the region of interest performed by Dawn includes the PIA22626 image, which contrary to previous images of the area appeared 180° vertically tilted on Dawn's gallery. This image was captured on July 6, 2018 from an altitude of about 36 miles (58 km), The sub-spacecraft position from which this image was taken is about 20.7° north latitude and 242.0° east longitude. This was the last image of the region published by NASA in Dawn's Ceres picture gallery (Fig. 7).

4. Discussion and conclusions

In a time where the search for techno-signatures is about to reshape SETI strategies and goals, a new perceptual and cognitive approach is needed. SETI does not depend exclusively on technology or algorithms but constitutes a cognitive task in itself, a human task limited as many others are by certain aspects dependent on our brain architecture, consciousness and neurophysiology. Here, we have presented some extraordinary examples that perhaps illustrate how our mind can easily

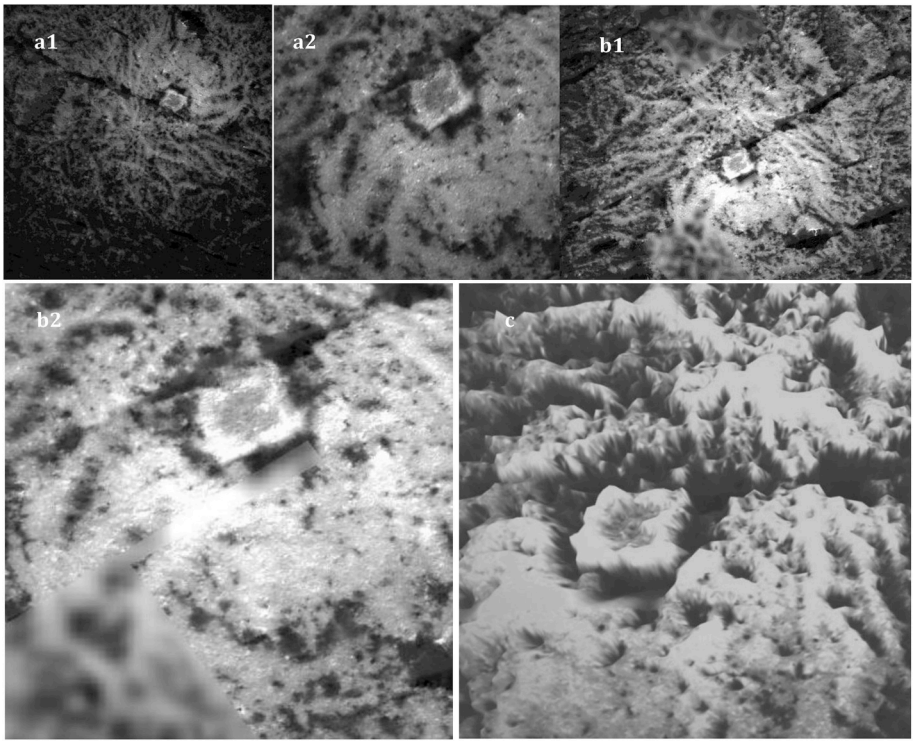


Fig. 7. Vinalia Faculae detail (VF-1): a1) 180° tilted image as it was published (July 16th) at 58 Km of altitude by Dawn. a2) Detail of a1 tilted of VF-1 to match the same orientation of the other existing images of the same area. b1) Mosaic image detail of VF-1 at 34 Km of altitude. b2) Image detail of b1. Credit. NASA JPL. Dawn. C) 3D elevation model of VF-1 performed by author using NormalMap freeware tool. Original image credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA/PSI.

experience cognitive dissonance when confronted with the unexpected. This leads us to two possible explanations: 1) cases such as Occator's VF-1 formation discussed here may represent examples where our perception may be insufficient because of biological and neurocognitive bias (participants were primed for squares not triangles), in a *Schiaparelli Mars canali*, or Face on Mars, fashion; 2) alternatively, some extraordinary characteristics have appeared in this particular VF-1 case. The presence of possible multiple, interrelated, geometries are certainly surprising but happened for both humans and AI, despite the plausible geological nature of this formation (salts, carbonates, etc.).

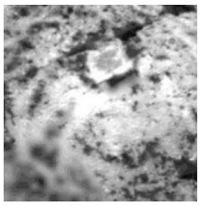
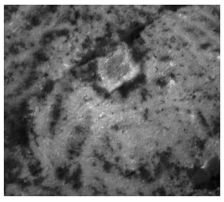
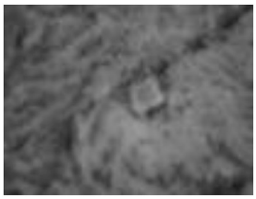
It has been widely discussed in the field of Psychology that inattention produces a failure of conscious perception. According to Mack [13] the unattended stimuli to which observers are functionally blind are perceptually as well as cognitively processed. They are analysed, and they consequently produce an implicit percept which is then encoded into an implicit memory store [13].

This percept, which has no presence in conscious awareness, is memory encoded and stored but there is no conscious access to it unless it is revealed through priming. If the implicit percept captures attention, it then becomes an explicit percept; that is, a conscious percept. If not, it remains an implicit memory. *"Implicit* or *unconscious* perception is fully processed, and is capable of capturing attention and will do so if it is highly meaningful to the observer when it is viewed under conditions of inattention" [13].

Cumulative implicit percepts and memories on the specific topic of the possible existence of other non-terrestrial intelligence may be stored for generations—probably based on unattended perceptions—building a not yet fully conscious construct or explicit percept until it becomes meaningful enough to become conscious or technological advances (AI) catalyse it. This type of cognitive phenomena, similar to how magicians manage to trick our attention, could be the effect of deliberately deceptive actions by advanced intelligences in unbalanced interactions, or may happen solely as a result of neurobiological/technological limitations of the species, or even a combination of both.

In our experiment, an AI CNN model trained to detect triangles and squares obtained similar results to those of humans, but critically different ones in the most controversial part. The results of our study raise some questions that are difficult to answer. First, if we suppose that the

Table 2
CNN AI computer vision model detection rate for the test image and two more of the same location at Vinalia Faculae, Occator crater, Ceres. Original image credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA/PSI.

Date and altitude	PIA21925 (VF-1)		PIA22626		PIA20653	
	 July 16th, 2018. 34 km		 July 6th, 2018. 58 km		 March 26th, 2018. 385 km	
	Triangle	Square	Triangle	Square	Triangle	Square
% detect	52.69	86.16	51.79	91.18	57.52	87.30

'Vinalia Faculae anomaly'(VF-1) is just a perceptual anomaly, AI did not help to disclose its real nature by acting in a way different from human bias but gave us a false positive. AI bias may arise as a possible problem in this field, as has been shown before in other scientific domains [14–17] This may be a concern for the future use of AI model applications in the SETI or exobiology searches. Second, AI offered marginal positive detection (triangle), creating a hard to solve and accept cognitive dissonance for VF-1 and its possible artificiality.

We did not include segmentation (exact localisation) processing in our analysis, but this could be an interesting line of research for the future analysis of VF-1. We believe that using AI systems may improve SETI performance and help overcome human bias in SETI tasks, but AI ethical aspects and human readiness for the possible outcome are yet-to-be defined factors requiring further research. It may be hard to accept that AI systems can surpass humans in more and more everyday activities including scientific tasks, not only because of our resulting secondary role in such activities but also because of the new concepts and realities that these AI systems may take us to. The question here is if we are ready to accept the outcome.

We do not have to completely abandon old strategies, but we can add new ones, including AI, to bring a different perspective to the search for evidence, but not only far away in cosmos but nearby too, because "life is fascinating: you only have to look at it through the right glasses" [18].

Funding

This study received partial funding support from the Department of Psychology of University of Cadiz. Spain. ID: 20DPPSOT00.

Declaration of competing interest

The author declares no competing interests.

Acknowledgment

Thanks Enrique Muñoz from BIT METRICS for his help with the CNN training and design.

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